Stellar Prices API — Milestone 2 Deliverable Evidence

- **Project:** Stellar Prices API
- **Team:** Rumble Fish

This document is the written companion to the Milestone 2 submission video. It maps every acceptance criterion from §9 of the technical design (*"Tranche 2 — Public API"*) to concrete evidence against the deployed production API: live URLs, runnable commands with their output, SQL a reviewer can execute, and code references.

Everything here is reproducible against the public deployment. The API base is `https://prices-api.sorobanscan.rumblefish.dev`, a REGIONAL custom domain mapped at the root, so there is no stage path in any URL. Key-gated routes need an `x-api-key` header; the key is available on request via the address on the submission form.

Two acceptance criteria are graded against amended wording, and both amendments are declared rather than assumed. AC 2's target is unchanged but no key in the account can sustain it; AC 3 names a response header this platform does not emit. Each is set out in full in `milestone-2-rfp-deviations.md`, which is part of this submission and should be read alongside this document.

This project runs as a second tenant on infrastructure the **Soroban Block Explorer** already operates — a shared AWS sub-account and a shared Hetzner ClickHouse cluster. The Soroban Block Explorer is abbreviated **SBE** below.

Evidence images are embedded inline in the published PDF and referenced from `screenshots/` in the Markdown source.

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1. Executive summary

Milestone 2 — **Public API** — is complete. All seven endpoint groups are deployed on a custom domain, key-gated, cached and validated, and **all six Tranche 2 acceptance criteria are met**. Two are graded against amended wording, and both amendments are declared in §4 with their reasoning rather than assumed.

AC	Criterion	Result
1	7 endpoint groups, schema-valid, ≥ 20 assets	**1032 checks pass, 0 fail, 0 skip**
2	100 req/s for 5 min, p95 < 200 ms, errors < 0.1 %	**p95 47.09 ms, 0 errors in 30,001 requests**
3	Cache confirmed within the TTL window	**hits 45-53 ms, misses 78-145 ms, no overlap**
4	VWAP verifiable against raw rows, ≥ 3 assets	**41 of 41 checks, 4 assets, worst delta 1.4e-11**
5	`earliest_data_available` ≤ 2022-01-01	**2015-11-18, reconciled four ways**

6	`timeframe=all` from ≥ Jan 2022, spot-checked	**2,042 points; spot-check on XLM and yBTC**

The four §9 work items with no numbered criterion are delivered (§6), as are the three items Milestone 1 explicitly deferred: the OpenAPI document through the gateway, the CloudWatch dashboard, and the API edge (§7).

What this document does not do is present a clean sweep. Three results carry limits that materially affect how they should be read, and each is stated where the evidence is rather than in a footnote.

- **The load-test pass is narrow.** It is cache-dominated, and an uncontended cache miss costs 170 to 240 ms — already at the Tranche 2 bar. Widening the pool to defeat the cache **took the production read path down for between 19 and 47 minutes**, and the root cause is not established.
- **AC 3's claim is weaker than the criterion asked for.** A header would be the cache asserting itself; latency is behaviour consistent with a cache. No `X-Cache` header exists on this platform and none can be produced honestly.
- **AC 6 substitutes its asset.** USDC's series is entirely peg-derived, carries no trades, and returns `1` through a real depeg, so it cannot fail the check and therefore cannot pass it. XLM and yBTC are spot-checked instead, over 28 dates rather than five.

§8 lists everything deliberately not claimed, with a destination for each row.

2. Deliverable definition

§9 of the technical design defines Tranche 2 as **Public API**, weeks 5 to 9. The work it names:

- The seven core endpoint groups, implemented and deployed: `GET /assets` (paginated, sortable, filterable), `GET /assets/{id}`, `GET /assets/{id}/price` (current price with a `sources` breakdown), `GET /assets/{id}/ohlcv` (timeframe and granularity parameters, with a `backfill_note` when history is partial), `POST /prices/batch`, `GET /oracles/{id}` (Reflector cross-reference), and `GET /backfill/status`.
- API Gateway response caching with per-endpoint TTLs, usage plans, API key issuance and throttling.
- The full VWAP formula wired into the current-price path (§5.5).
- Outlier detection, excluding sources that deviate beyond a configurable percentage from the inter-source median.
- Aquarius pool metadata integration, so Aquarius appears as a named source in the VWAP breakdown.
- Input validation: asset identifier format enforced, parameter ranges validated, `400` on invalid input.

The backfill milestone for the tranche is SDEX history covering approximately January 2022 to present, written directly to Hetzner over mTLS per ADR 0009, with the covered range visible through `GET /backfill/status`.

3. Architecture

Unchanged in shape from Milestone 1, extended at the API tier. The full description, including the component table and the data-flow diagram, is `docs/prices-api-general-overview.md` §2 and §3.

What Tranche 2 added on top of the Milestone 1 platform:

Layer	Addition
API Gateway	Per-method response cache (0.5 GB), usage plans, key issuance, two-level throttling, CORS preflight on all seven `/v1` routes, REGIONAL custom domain
Lambda (axum)	The six read route groups beyond `/backfill/status`, input validation, the OpenAPI document served from the code
ClickHouse	`current_prices` materialised with the VWAP breakdown, `usd_rate` for measured USD conversion, the `method` provenance column

The API is a single Rust axum Lambda behind API Gateway (ADR 0008), with no VPC and no NAT gateway. Reads go over the public internet to Caddy on Hetzner with mutual TLS.

4. Deviations from the approved wording

Three places where the delivered system departs from the literal wording of the RFP or of the Tranche 2 criteria. Each is set out in full, with its measurements and its reasoning, in `milestone-2-rfp-deviations.md`. That document is part of this submission; the rows below are pointers, not summaries.

#	Wording says	We deliver	Kind
1	`Current Price (float USD)`	a decimal **string**, so 14 fractional digits survive transport	defended
2	AC 3: *"...return `X-Cache: Hit` header"*	a reworded criterion graded on latency and deployed cache configuration; **no such header exists on this platform**	disclosed
3	AC 6: USDC 1d candles spot-checked	XLM and yBTC over 28 dates; USDC excluded, reason stated	disclosed

AC 2 also carries an in-place amendment, already recorded in §12 of the design document since before this submission: the 100 req/s target is unchanged, but since task 0157 no key on the default plan can sustain it, so the run needs a usage plan created for it and the report must say which plan the key was on.

Declaring in-tranche refinements in the package itself is the discipline Milestone 1 was accepted on. Its form answers carry an *"In-tranche scope refinements"* section, and its evidence document carries a *"what is deliberately not claimed"* list.

5. Acceptance-criteria evidence

AC 1 — All 7 endpoint groups return correct, schema-valid responses for ≥ 20 major assets
Verdict: met. 1032 checks pass, 0 fail, 0 skip, against the deployed production API on **2026-09-07 at 10:40 UTC**, and again at **11:09 UTC** with an identical verdict.

The evidence is a scripted, re-runnable conformance suite (`tools/scripts/conformance-0120.mjs`, task `0120`). It exercises all seven route groups for the twenty assets fixed in `tools/scripts/conformance-assets.json`, a list derived from production volume rank on 2026-08-19 and deliberately covering all three identifier forms: the native asset, code-with-issuer, and a Soroban contract address.

Every response is validated against the API's own live OpenAPI document, fetched from `/api-docs-json` at run time rather than from a checked-in copy, so the API is measured against the contract it actually publishes. Error responses are validated too. **There has been no schema failure in any run to date**; the entire failure surface was ever only the correctness layer above the schema.

On top of the schema sit the assertions a schema cannot express: OHLC ordering on every priced bucket, bucket alignment and strictly increasing timestamps, cursor pagination proven exhaustive and duplicate-free, the batch endpoint agreeing with the per-asset endpoint at a matching snapshot, decimal strings that survive the JSON round trip, and documented sentinels asserted as sentinels.

Reproduce it

```
# API_KEY and BASE_URL from the environment or .env.local at the repo root
npm run conformance:0120 && echo "TRANCHE 2 AC 1: PASS"
```

The suite paces itself at 1.1 s per request, matching the free plan's 1 req/s sustained rate, and drives no load. It writes a timestamped JSON report of every individual check.

The trajectory, and what it does and does not show

run	pass	fail	skip
2026-08-19, first pass	752	55	0

2026-08-25	847	13	11
2026-09-02	870	16	0
2026-09-07, 10:40	**1032**	**0**	**0**
2026-09-07, 11:09	**1032**	**0**	**0**

🔑 **Two things must be read alongside that table, and we would rather state them than have them noticed.**

The check count rose, 886 to 1032. Nothing was relaxed to reach zero. Every change on 2026-09-07 *added* assertions, and each one replaced an assumption with a measurement.

No production code changed on 2026-09-07. Every failure that disappeared that day was a defect in the test, not a fix to the API. Three of them were the same mistake in different places: an assertion that encoded a market state rather than the contract. The suite treated a *decided* sentinel as a pending defect; it failed buckets that ADR 0011 §5 returns unpriced on purpose; and it called all twenty fixture assets liquid when two of them had gone quiet since the list was ranked three weeks earlier.

Why the report is citable rather than a snapshot

An acceptance report whose result moves with a background worker cannot be evidence. The suite used to have exactly that property: three assets failed at 13:26 and passed at 13:38 on 2026-08-27 with unchanged code, because whether a bucket carried a USD price depended on how far enrichment had advanced.

The two runs above were taken **29 minutes apart** and diffed check by check:

checks compared	1,032
identical verdicts	**1,032**
verdicts changed	**0**
underlying details that moved	**53**

The moving details are what make the identical verdicts mean something. Four assets advanced at the tip during the gap — `native`, `EURC`, `AQUA` and `BTC` each went from `0 of 169 buckets unpriced` to `1 of 168`. **Under the previous assertion those four flip from pass to fail.**

Stated limits

- **The reports are gitignored as regenerable.** The citable artefact is the figures above plus the one-command reproduction, not a committed file.
- **The pagination walk moves between runs** — 19 pages and 3,725 distinct assets here, against 18 / 3,567 and 20 / 3,880 earlier. Exhaustive and duplicate-free each time; the traded population itself is what moves. Owned by task 0261.
- **One assertion is deliberately weaker than the rest.** A candle's timestamp is its bucket *start*, so "did this asset trade in the last 24 hours" cannot be answered exactly from candles alone. The liquidity test is three-valued: certainly inside the window, certainly outside, or a residual band one bucket wide that accepts either documented response. Claiming to resolve that band would assert something the data does not carry.
- **The fixture list was not re-derived** to today's volume ranking. Doing so would break comparability with the four earlier runs.
- USDT is excluded from the fixture, a known defect recorded as task 172.
- Three defects found during the pass were spawned rather than fixed: tasks 0210, 0211 and 0261.
- This is **not** the Tranche 3 deliverable *"integration test suite runs in CI"*. It is an operator-run acceptance check.

AC 2 — Load test: 100 req/s for 5 minutes, p95 < 200 ms, error rate < 0.1 %

Verdict: met, on the amended wording. p95 = 47.09 ms against a 200 ms bar, and 0 errors in 30,001 requests against a 0.1 % bar. Measured 2026-09-03, 06:04–06:16 UTC, with k6 v2.2.0 (task 0121, `prices-api-load-test-100rps.md`).

The amendment, already recorded in §12 before this submission: the target is unchanged, but since task 0157 the default plan is 1 req/s with a 100,000 monthly quota, so no ordinary key can sustain the run. A single 5-minute pass at 100 req/s is 30,000 requests, nearly a third of a month's allowance. The run therefore used a purpose-made plan, `prices-production-loadtest-plan` at 150 req/s, burst 300 — and the criterion asks the report to say so, which it does.

regime	pool	p50	p95	p99	achieved rate	non-200	error rate
1 — cache	1 asset	45.15	47.75	175.95	99.87 req/s	0	0.00 %
2 — the AC scenario	**18 assets**	**45.04**	**47.09**	**107.53**	**99.85 req/s**	**0**	**0.00 %**
3 — wide	4,301 assets	64.97	76.91	183.16	94.28 req/s	28,314	94.38 %

Latencies in milliseconds, `phase:main`. Regime 2 is the criterion's scenario. All four k6 thresholds held and the process exited 0, with zero dropped iterations.

Reproduce it

```
S='avg,min,med,max,p(50),p(90),p(95),p(99)'
k6 run packages/prices-api/loadtest/price_load.js \
  -e BASE_URL=https://prices-api.sorobanscan.rumblefish.dev -e API_KEY="$API_KEY" \
  --summary-trend-stats="$S" --summary-export=loadtest-ac.json; echo "exit=$?"
```

`exit=0` means every threshold held, and that is what the report cites: in the exported JSON a threshold recorded as `false` means *passed*, which is too easy to misread. The trend-stats flag is required, because k6 stops at p95 by default. Passing `-e ASSET=native` would pin a single asset and measure the gateway cache instead of the criterion's scenario. Raw exports for all three regimes are in `docs/loadtest-results/`.

🔴 **The pass is narrow, and quoting it alone would mislead**

Regime 2 is cache-dominated. With an 18-asset pool and a 10-second TTL the hit rate cannot fall below 98.20 %, so 47.09 ms is substantially the gateway cache answering, not the data path. Regimes 1 and 2 are statistically indistinguishable at p95.

An uncontended cache miss costs roughly 170 to 240 ms. That is already at the Tranche 2 bar and about twice the Tranche 3 bar of 100 ms. The pass rests on the cache concealing a data path that is, on today's measurements, too slow for Tranche 3. **p99 already exceeds 100 ms in both passing regimes.**

🔴 **Regime 3 was an incident, not merely a failed run.** Widening the pool to 4,301 assets drove the hit rate to approximately zero and took the production read path down. 28,314 of 30,001 requests returned `500 db_error`. Single sequential requests were still failing minutes later. **The outage was at least 19 and at most 47 minutes**, and the width of that range is our own process failure: the liveness probe died after its sixteenth check and the restart failed silently. The root cause is not established and is open as task 0260.

Regime 3's latency column is therefore **not a latency measurement** and must not be read as one, since 94 % of it is the speed of returning an error.

A standing safety rule follows from this, and it governs the cache evidence below: do not verify behaviour by driving cache misses at rate.

Deferred

- **Cache-hit and cache-miss percentiles reported separately** — deferred to task 0260. Hits are measured. Misses are not obtainable by this method, because the system stops serving before a miss percentile can be sampled at 100 req/s.
- **Cold-start incidence and ClickHouse-side query time** — deferred to task 0260, blocked on production CloudWatch access rather than on the run.
- The 20-asset pool ran as **18**: two assets returned 404 during setup and were excluded before measurement. Which assets are unservable drifts day to day, a data-freshness property rather than a fixed defect list.

AC 3 — Cache confirmed within the TTL window

This criterion is graded against amended wording. Read `milestone-2-rfp-deviations.md` §2 for the full argument; it is not reproduced here.

The criterion as written asks for consecutive identical requests inside the TTL window to return an `X-Cache: Hit` header. **This API emits no `X-Cache` header on any route**, and cannot be made to emit a truthful one. That is not a setting left switched off: API Gateway's stage cache has no hit-or-miss header feature.

It is graded instead against this wording:

> *"Cache confirmed: consecutive identical requests within the TTL window are served from the API Gateway stage cache, and a request after the window is not. Demonstrated by response latency, which separates cleanly, and by the deployed per-method cache configuration."*

🔑 **The claim weakens, and we state it in those words rather than blur it. A header would be the cache asserting itself. Latency is behaviour consistent with a cache.** That is a weaker form of proof than the criterion asked for.

Verdict on the amended wording: met. Measured 2026-09-03.

The cache works, and it expires when it says it does

`/price`, declared 10-second TTL, same URL throughout. Server time only, so the TLS handshake a fresh client pays is excluded.

request	expected	server time
first ask	miss	144.7 ms
+2 s	hit	53.3 ms
+4 s	hit	45.1 ms
+6 s	hit	47.7 ms
+13 s	**miss, expired**	**139.7 ms**
+15 s	hit	49.5 ms

Hits fall between 45 and 53 ms, misses between 78 and 145 ms, and the two ranges do not overlap. Expiry is demonstrated on both TTL tiers: `/v1/assets` at a 60-second TTL was still hot at +32 s and expired at +64 s. Both refilled immediately. These hit figures independently reproduce the load test's 45 to 47 ms, from a different tool on a different day.

All six key-gated cached routes show the pattern. The deployed per-method configuration was read straight off the production stage, and three surfaces agree: the deployed stage, the CDK constant, and the handler's own cache-control tiers. `x-api-key` appears in no cache key, so the cache is correctly shared across callers for identical public data.

Reproduce it

```
BASE=https://prices-api.sorobanscan.rumblefish.dev
srv() { curl -s -o /dev/null -H "x-api-key: $API_KEY" \
  -w '%{time_starttransfer} %{time_appconnect}' "$1" |
  awk '{printf "%.1f ms\n", ($1-$2)*1000}'; }

U=$(date +%s)                                    # any unused value forces a miss
P="$BASE/v1/assets/native/price?min_volume_usd=$U"
srv "$P"; sleep 2; srv "$P"; sleep 2; srv "$P"   # miss, hit, hit
sleep 9;  srv "$P"                               # miss — TTL expired
sleep 2;  srv "$P"                               # hit — refilled
```

Because `min_volume_usd` is part of the `/price` cache key, a guaranteed miss costs one request. No cold-asset hunting, no waiting, and **no load generation**.

Why the header cannot honestly be added

A header written by our own handler would be **actively wrong rather than merely absent**. API Gateway replays a cached response byte for byte and the Lambda runs only on a miss, so the header would freeze at miss time and report `Miss` on every genuine hit. Measured, not assumed: the response body hash held at `4694801b` across every hit and changed to `039f54a6` only when the entry expired.

CloudFront, the only architecture that emits a truthful hit-or-miss header, writes `X-Cache: Hit from cloudfront` — so the literal wording would still be unmet after three to five days of edge work and a DNS migration. Cost is explicitly not the argument; the two are within a couple of dollars a month.

The decision is recorded as **ADR 0012**, accepted and ratified by the team lead, with all three alternatives closed with reasons and an explicit list of what would justify revisiting it.

🔴 Stated limits — the evidence is not uniformly strong

The gap between a hit and a miss tracks how expensive the underlying query is, so the method discriminates unevenly across routes.

route	miss / hit	daylight
`/ohlcv`	2.7×	120 ms
`/price`	~2.9×	~95 ms
`/backfill/status`	**1.4×**	**19.5 ms**

19.5 ms is real and repeatable, but close enough to ordinary variance that a single pair of requests would not settle it on that route. There it is carried by the deployed configuration rather than by latency. **A header would not have had this property**, and presenting the six routes as uniformly strong would overstate the evidence.

- **`/api-docs-json` is confirmed configured** at a 3600-second TTL but deliberately **not** claimed as demonstrated: its entry clears only on a deploy flush, so no miss can be induced to compare against.
- **The hit rate under load is derived, never observed** — there is no header to count. The bounds are ≥ 99.90 %, ≥ 98.20 % for the criterion's scenario, and approximately 0 % for the wide pool. **The derivation holds only because the load script sends no query string.** A client varying `min_volume_usd` gets one cache entry per value and none of these figures apply.
- `POST /prices/batch` and `/health` are uncached, established from the deployed stage configuration. `/health` timings are not offered as evidence, because it is a gateway mock and fast either way.

AC 4 — VWAP verifiable against raw `price_ohlcv` rows for ≥ 3 assets

Verdict: met, on four assets rather than three. 41 of 41 checks reconciled. Measured against a pinned tick, `T = 2026-08-26 13:22:00 UTC`, over the preceding 24 hours (task 0123).

The method is a from-scratch recomputation. The published `current_prices` row is compared against **29,108 raw `price_ohlcv_1m FINAL` rows** captured directly from ClickHouse, re-aggregated in **plain Python with no ClickHouse involvement**, so the check is independent of the materialised view's own SQL rather than a restatement of it.

asset	sources	VWAP relative delta	volume delta
XLM	4	`5.1e-14`	exact, `0E-14`
EURC	4	`8.1e-15`	exact
BTC	2	`1.9e-16`	exact
AQUA	3, guard cuts to 2	`1.4e-11`	exact
SCOP (control)	2, filter cuts to 1	`0`	exact
USDCAllow (control)	1	`0`	exact

Four of these are multi-source and count toward the criterion's bar of three: XLM, EURC, BTC and AQUA. The stated tolerance on `vwap_24h` was a relative difference of 1e-9; the worst observed was **1.4e-11**, roughly seventy times tighter. `volume_24h_usd` is a plain sum and was asserted **exactly equal**, which held on all six.

The exclusions are attributed, not just tolerated. AQUA's third source was dropped by the liveness guard, being stale since 07:53. SCOP's second was dropped by the population filter, having 25 candles none of which were priced. Each exclusion is explained by a named rule.

A second run at tick `2026-08-26 14:17:00Z` carried the check through the public API: **13 of 13 fields were Decimal-value-exact and string-typed** in the JSON response, so nothing is lost between ClickHouse and the wire.

Reproduce it

```
cd lore/1-tasks/archive/0123_TEST_vwap-reconciliation-against-raw-ohlcv
python3 benchmark/reconcile.py benchmark/current.csv \
  <(gunzip -c benchmark/raw-T1322.csv.gz)
```

The capture query is `benchmark/q-raw.sql`, and the public-API leg is `GET /v1/assets/native/price`.

Stated limits

- 🔑 **Ties across quote legs are common, not exotic.** Four of the six assets had two or more rows sharing the newest priced timestamp, so "the latest priced close" is a **set**, and which member the view picks is not contractual. Prices are therefore asserted as set membership. This matters because the first draft of the reconciliation picked an arbitrary tie member and produced a **false mismatch of 3.0e-04 on AQUA**.
- **ETH was deliberately excluded at selection**, because its two venues shared a newest-priced timestamp and disagreed on price. Any assertion on it would have been flaky, and saying so is better than quietly dropping it.
- **The exclusion sets must be re-derived per run.** Between selection and capture, one venue revived and the guard case moved from one asset to another.
- The `min_volume_usd` threshold did not exist when this ran, so no threshold exclusions appear. It shipped afterwards as task 0118, covered in §6.
- Side product, carried into outlier tuning: the worst per-venue deviation from the inter-source median observed here was **0.824 %**, against a configured band of 20 %. The outlier mask excluded nothing, and that headroom is the starting evidence for task 0217.

AC 5 — `GET /backfill/status` shows `earliest_data_available` ≤ 2022-01-01

Verdict: met, by six years. The criterion asks for 2022-01-01. The store reaches **2015-11-18**. Measured 2026-09-04 (`prices-api-backfill-depth-verification.md`, task 0127).

The figure is reconciled four ways, deliberately, because the endpoint reports a stored value rather than querying the candles:

view	oldest SDEX data
`GET /backfill/status`, `sdex.earliest_data_available`	`2015-11-18T03:47:00Z`
the stored `backfill_progress` row	`2015-11-18 03:47:00`
`min(timestamp)` on `price_ohlcv_1d`	`2015-11-18 00:00:00`
oldest active partition, all seven candle tiers	`201511`

Depth alone is not coverage, so continuity was checked too. Every month from 2022-01 to 2026-09 carries SDEX candles on **every calendar day** — all 57 months, leap day included. In the criterion's own year there are **4,048,196 daily SDEX candles across 82,096 assets**. Two visibly lower-density months were examined day by day and showed no empty days and no cliff: breadth of assets varying, not data loss.

Reproduce it

```
curl -sS -H "x-api-key: $API_KEY" \
  https://prices-api.sorobanscan.rumblefish.dev/v1/backfill/status | jq .sdex
```

ClickHouse read-only `SELECT`s against `prices.price_ohlcv_1d` and `prices.backfill_progress` reproduce the other three views. A reviewer can be issued a short-lived read-only client certificate on request.

Stated limits

- 🔑 **The stored value is a monotonic high-water mark and cannot correct itself downward.** An overstatement would be permanent and invisible. It does not overstate here, but that property is exactly why the reconciliation above exists rather than trusting the endpoint alone.
- ⚠️ **The endpoint contradicted itself on this very field and was fixed during the tranche.** The SDEX stream reported `completed` alongside `progress_pct: 0` and a ledger count implying nothing had been done, because the archive walk runs *downward* toward genesis while the progress arithmetic assumed upward. Fixed and merged as PR #283.
- ⚠️ **A neighbouring field in the same payload overstates.** The AMM stream claims data from 2024-02-20 while the first actual AMM candle is 2024-03-08. That is 17 days claimed at no granularity. Deferred as task 0263 and task 0264 rather than quietly corrected.
- 🔑 **The `sdex.last_push_at` freshness criterion no longer applies**, and this package says so rather than quoting a number. The value is 2026-08-11 on a stream that reached `completed` on 2026-07-27. A corollary for operations: the Tranche 1 alarm watching that column will fire forever unless it is gated on the stream's status.
- `backfill_note` is correctly **absent** for the same reason, its precondition having lapsed. Both branches are covered by integration tests.

AC 6 — `timeframe=all` on USDC returns 1d candles from ≥ January 2022, spot-checked

Verdict: the literal half is met. The spot-check uses different assets, and that substitution is declared, not slipped in.

`GET /v1/assets/USDC:GA5Z…/ohlcv?timeframe=all` returns **2,042 daily points spanning 2021-02-01 to 2026-09-04**, with no gaps. That satisfies the criterion's first clause as written.

🔴 Why USDC is excluded from the correctness half

The criterion's second clause asks for candles *verifiable against known USDC price history*. USDC cannot serve that purpose here, and the reason is worth stating precisely.

	USDC	`native` (control)
points returned	2,042	2,414

points with `trade_count > 0`	**0**	**2,414, all of them**
marked `derived`	all 2,042	—
distinct closing values	**177, of which 1,865 are exactly `1`**	real market values

Not one of those points has a trade behind it. The series is peg-derived. The falsifying case is the obvious one: on **2023-03-11**, when USDC broke its peg to roughly $0.87 after the Silicon Valley Bank failure, we return exactly 1. XLM on the same day moved across **36,208 trades**, so the underlying data exists; it is simply not used for USDC.

A series that cannot report a depeg is repeating an assumption rather than reporting a price. It would pass this criterion by construction, which is why it cannot meaningfully pass it. The durable fix is scheduled for Milestone 3 as task 0265.

What is delivered instead

XLM and yBTC, over 28 dates each rather than the five asked for, against Binance daily klines. Binance is public, needs no key, has history predating 2022, and shares no price source with us.

asset	median absolute deviation	within 5 %	worst
XLM	**0.06 %**	**27 of 28**	16.2 %
yBTC	**0.48 %**	**27 of 28**	14.8 %

Five reviewer-style dates on XLM: 2022-01-03 at −0.19 %, 2022-06-15 at −0.42 %, 2024-07-01 at −0.22 %, 2026-06-15 at +0.01 %, and 2023-03-11 at −25.35 %.

Reproduce it

```
curl -sS -H "x-api-key: $API_KEY" \
  "https://prices-api.sorobanscan.rumblefish.dev/v1/assets/native/ohlcv?timeframe=all&granularity=1
d"

curl -sS "https://data-api.binance.vision/api/v3/klines?symbol=XLMUSDT&interval=1d&startTime=<UTC-
midnight-ms>&limit=1"
# field 4 is the close
```

🔴 Stated limits — the outliers are not noise

Both assets dislocate by the same ratio on the same two dates. On 2023-03-11 the ratios are 0.746 and 0.744; on 2023-03-15 they are 0.838 and 0.852. Every neighbouring day is clean. Two unrelated assets moving by an identical factor indicates a **shared cause, and the mechanism is not yet established.** It is tracked as task 0266. The obvious peg explanation is **refuted**: it would require USDC at about $1.34, and USDC fell.

- **The reference is not neutral on exactly those dates.** Binance quotes against USDT, which itself traded at a premium during the same stress window.
- **yBTC is thin.** A few hundred trades a day means a single off-market close moves the figure, which is what one 2023 date shows while XLM is clean beside it.
- ⚠️ **USD coverage across the whole estate is thin, and thinnest where it is oldest.** `close_usd` is zero on **36.20 % of 2022 daily SDEX candles**, 13.32 % of 2021, and effectively all of 2015 to
  2020. **The spot-check assets were chosen from the priced majority, deliberately**, and that choice is disclosed here rather

  than left for a reviewer to discover.

6. Work items without a numbered criterion

§9 names four pieces of Tranche 2 work that carry no numbered acceptance criterion. All four are delivered.

6.1 Full VWAP formula, with a minimum-volume source threshold

§5.5 defines the weighted price as the volume-weighted mean of per-source prices, including only sources above a configurable minimum volume.

🔑 **One correction to §9's wording, stated rather than glossed.** The formula does not run in a "Current Price Updater Lambda". It runs in `prices.mv_current_prices`, a ClickHouse materialised view refreshing every minute, which is the sole writer of `current_prices`. That substitution was made early and removed two Lambdas from the design.

The `sources` breakdown and the weighting arrived with task 0072. Measured refresh cost on production: **176 to 211 ms**, reading 1.69 M rows to write about 3,000, comfortably inside the one-minute refresh interval.

The threshold arrived with task 0118, and **a production measurement taken before merge reversed its design.** Applied unconditionally, a $100 floor would have blanked the VWAP and sources on **2,960 of 3,068 priced assets, or 96.5 %** — because 82 % of venues carry a dollar a day or less. The threshold was therefore made **conditional**: a below-threshold source is dropped only when a funded source survives. An explicit `?min_volume_usd=` override remains strict, and that asymmetry is deliberate.

Live evidence, all four responses pinned to one refresh tick on XLM:

request	sources returned	`vwap_24h`
no parameter	aquarius, sdex, soroswap	`0.18374529364442`
`?min_volume_usd=100`	aquarius, sdex, soroswap	`0.18374529364442`
`?min_volume_usd=5000`	aquarius, sdex	`0.18374945623882`
`?min_volume_usd=200000`	aquarius	`0.18383237183385`

```
curl -sS -H "x-api-key: $API_KEY" \
  "https://prices-api.sorobanscan.rumblefish.dev/v1/assets/native/price?min_volume_usd=5000"
```

⚠️ **A production defect was found by verifying after deploy, and fixed in the same change.** API Gateway does not key its cache on the query string, only on declared cache-key parameters. One filtered request therefore **poisoned the default response for every other caller** for the length of the TTL window: a narrowed `sources` came back to a caller who had sent no parameter at all. Fixed by declaring the parameter on both routes. The planning note for that task had asserted the opposite, and the measurement is what caught it.

6.2 Outlier detection

Sources deviating beyond a configurable percentage from the inter-source median are excluded from the VWAP. Configured at **20 %**, and it **arms only at three or more sources**.

🔑 **Three limits that are properties of the design, not defects, and are better stated than discovered.**

The three-source guard is load-bearing. With two sources the median is the midpoint, so any threshold either keeps both or drops both; with one it is that source. Below three sources the filter is a no-op by construction.

At three or more it can clear every source. The median interpolates on an even count, so four values of 1, 1, 3, 3 give a median of 2, every element deviates by 50 %, and the source set empties while the headline price still publishes. **An asset can therefore carry a price with no sources beside it.** Whether interpolation is the right median here is open as task 0238.

The headline price is not outlier-filtered — only the VWAP is. §7 scopes outlier detection to the VWAP, so this is a deliberate gap rather than an oversight, and the design document now states the asymmetry instead of hiding it. The consequence is real: the headline price can come from a manipulated venue while the VWAP beside it is protected. The decision on whether to filter it moved to task 0217.

The 20 % band is **a starting value, deliberately loose**. The tuning evidence is AC 4's reconciliation, where the worst observed venue deviation was 0.824 %, some twenty-four times inside the band, and the mask excluded nothing.

6.3 Aquarius as a named source

Delivered, and visible in the public response rather than only in the pipeline:

```
"sources": {
  "aquarius": { "price": "0.18206086877515", "volume_24h": "302077.47208850593463" },
  "phoenix":  { ... }, "sdex": { ... }, "soroswap": { ... }
}
```

That is a live payload for XLM. Aquarius appears with a real price and real 24-hour volume, and at a `?min_volume_usd=200000` filter it is the last source standing.

```
curl -sS -H "x-api-key: $API_KEY" \
  https://prices-api.sorobanscan.rumblefish.dev/v1/assets/native/price | jq '.sources | keys'
```

⚠️ **Aquarius coverage is not complete, and the gap is disclosed.** Twenty mainnet Aquarius *concentrated* pools are held back from the seeded pool registry, because the extractor is written for constant-product and stableswap pools only. Their swap events carry amounts without an identifiable token ordering, so decoding them is still open as **task 0080**. Any volume those pools carry is absent from the source's 24-hour figure, which can in turn push it under the threshold in §6.1.

6.4 Input validation

Delivered as task 0119: identifier format enforced, parameter ranges validated, `400` on invalid input, with **about 50 negative tests running in CI without a database** plus 21 integration tests against live ClickHouse.

The structural proof is worth naming, because it is stronger than the test count. The negative tests run against a handler state that **panics on any database access**, so every clean `400` also proves that no query was issued before the rejection.

Caps: page size 1 to 200, search strings to 64 bytes, cursors to 256 characters, batches to 100 elements behind a 16 KB body limit, and any window-times-granularity combination above 5,000 buckets rejected.

⚠️ **That last cap is a consumer-visible breaking change** and is listed here for that reason. A request spanning 30 days at one-minute granularity is 43,200 buckets and now returns `400` where it previously returned the newest 5,000 silently. Two smaller semantic shifts are recorded with it: naive timestamps are now pinned to UTC in the handler rather than inheriting the database server's timezone, and a far-future end bound now anchors the window instead of behaving as no bound.

A known limitation is stated rather than left latent: the pagination cursor does not record which sort produced it, so switching sort order mid-walk yields a wrong page. The fix is deferred to task 0206.

7. Milestone 1 deferrals, now delivered

Milestone 1's evidence document listed what it deliberately did not claim. Three of those rows were Tranche 2 scope. All three are now delivered.

7.1 The OpenAPI specification is served through the gateway

Delivered. `GET /api-docs-json` is mapped as a keyless, cached route on the production API, so a reviewer needs no credential to read the contract.

```
curl -sS https://prices-api.sorobanscan.rumblefish.dev/api-docs-json \
  | jq '{openapi, servers, paths: (.paths | keys)}'
```

The document is **OpenAPI 3.1.0**, generated from the axum routes rather than maintained by hand, so it cannot drift from the code. It is also rendered as the portal's API reference. Three artifact-derived guards run in CI: the full test suite, a Redocly `recommended-strict` lint that reports **zero errors and zero warnings**, and a route-parity check confirming the gateway and the document agree, which was itself confirmed to fail in both drift directions.

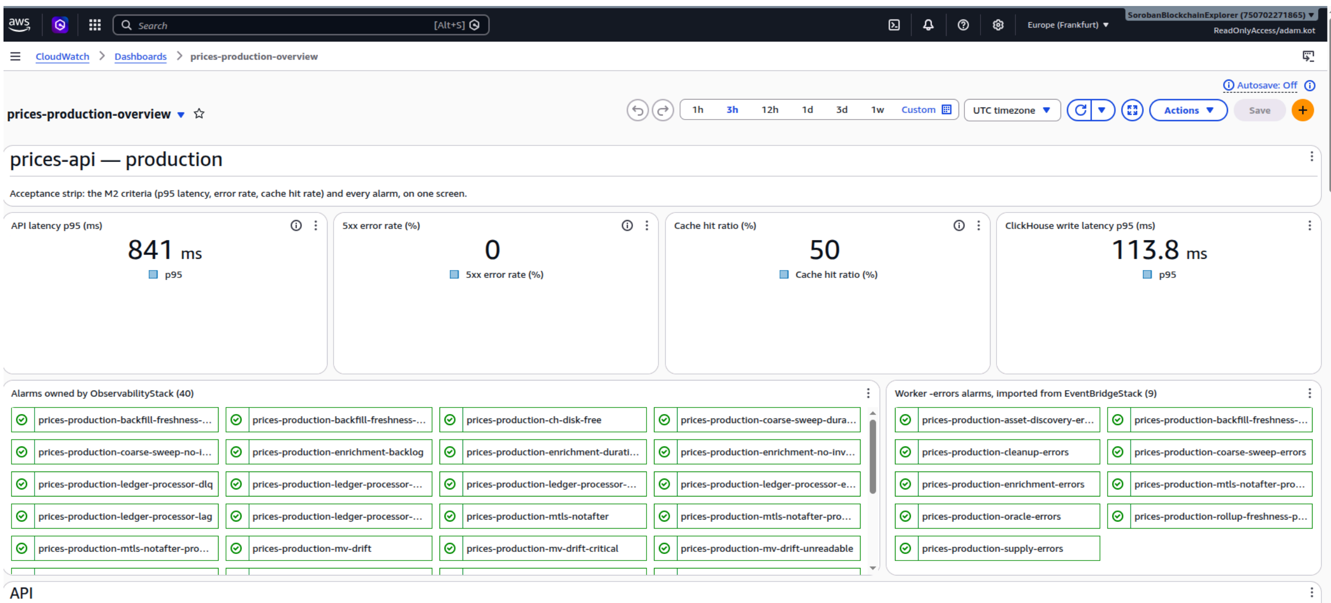
⚠️ **A second validator disagrees, and the disagreement is disclosed rather than hidden.** The Tranche 3 criterion names `openapi-validator`. The gate we ship is Redocly, which the document passes cleanly. IBM's validator, which owns that name on npm, reported 13 findings on the same bytes; eight were fixed and **five remain deliberate**, with the full accounting recorded in the task. We are flagging the interpretation rather than editing the criterion.

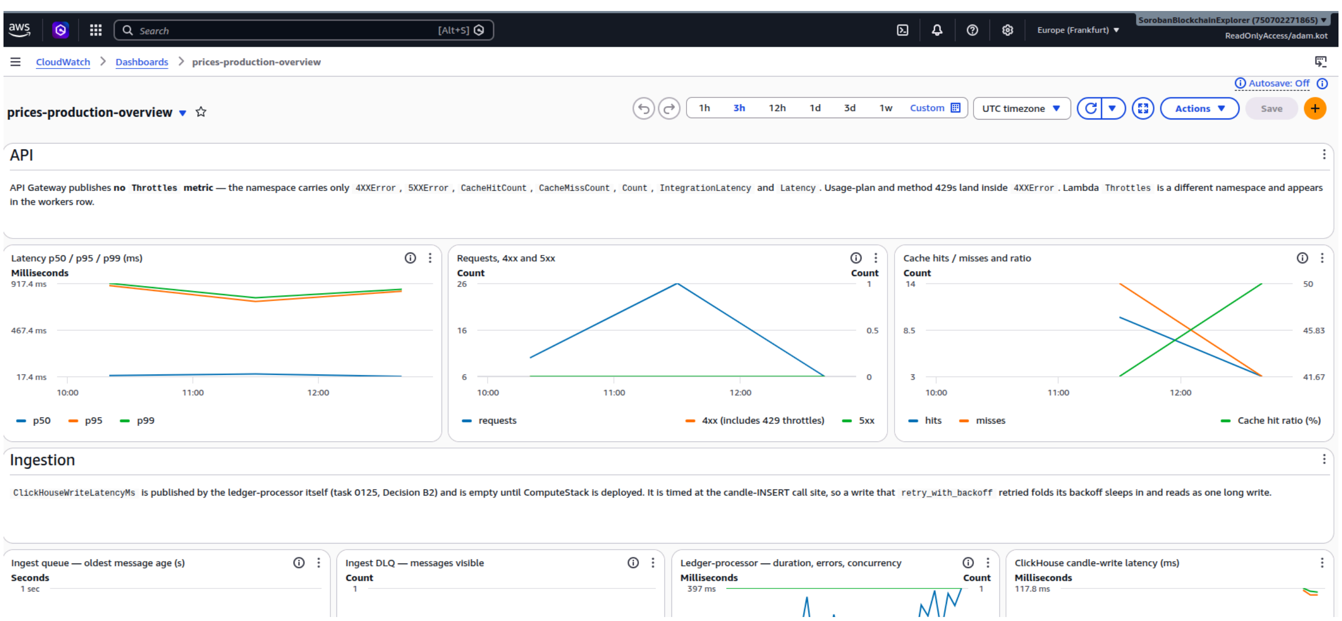
The licence field is currently emitted empty, pending an open decision recorded as task 0155.

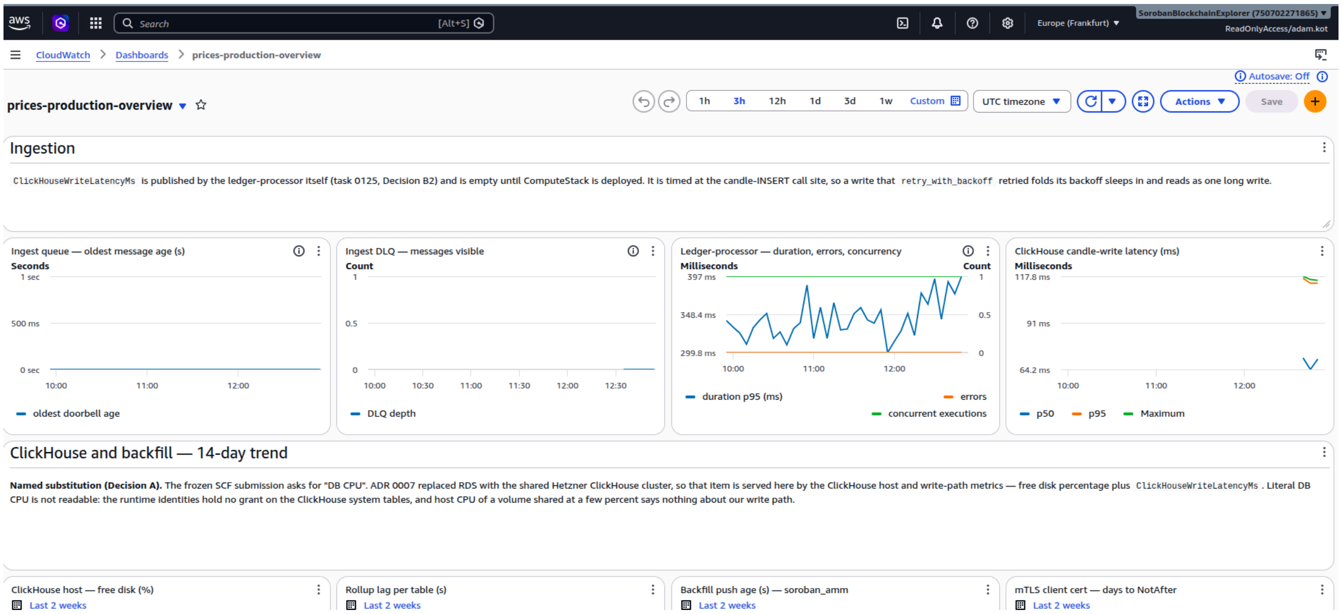
7.2 The CloudWatch dashboard has real data widgets

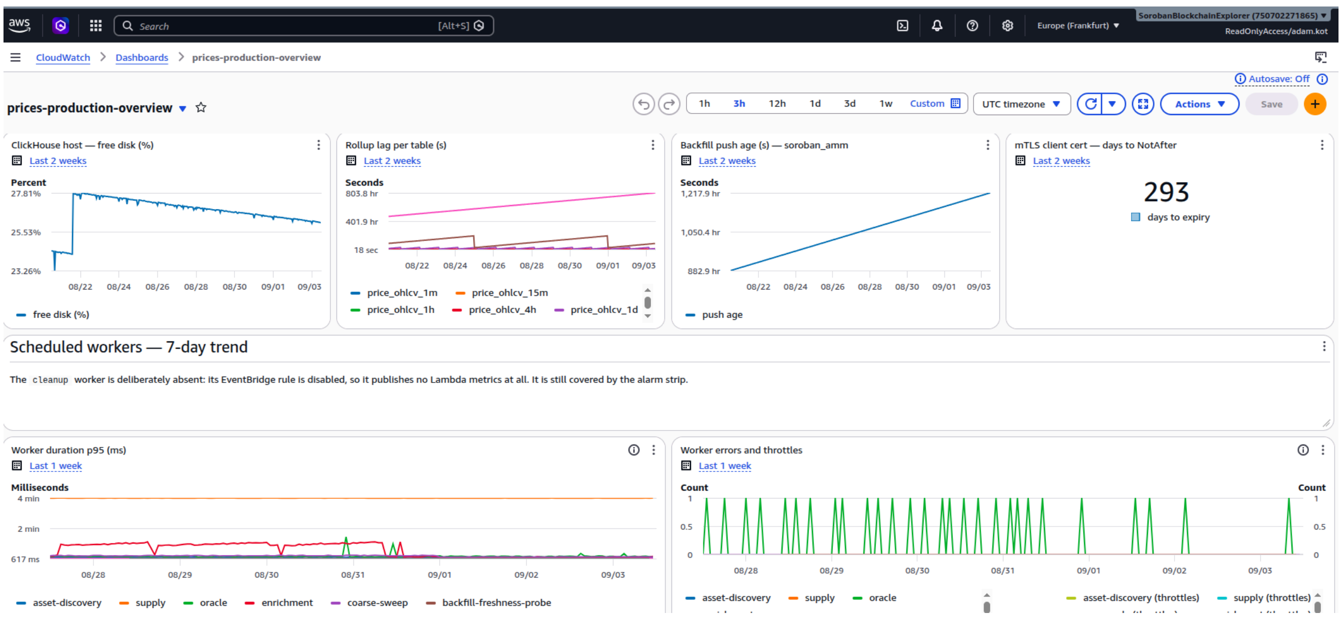
Delivered. `prices-production-overview` was a scaffold with no data widgets at Milestone 1. It now carries six rows: an acceptance strip of API p95 latency, 5xx rate, cache-hit ratio and ClickHouse write latency, then the alarm strip, then API, ingestion, ClickHouse and backfill, workers, and enrichment and oracle panels.

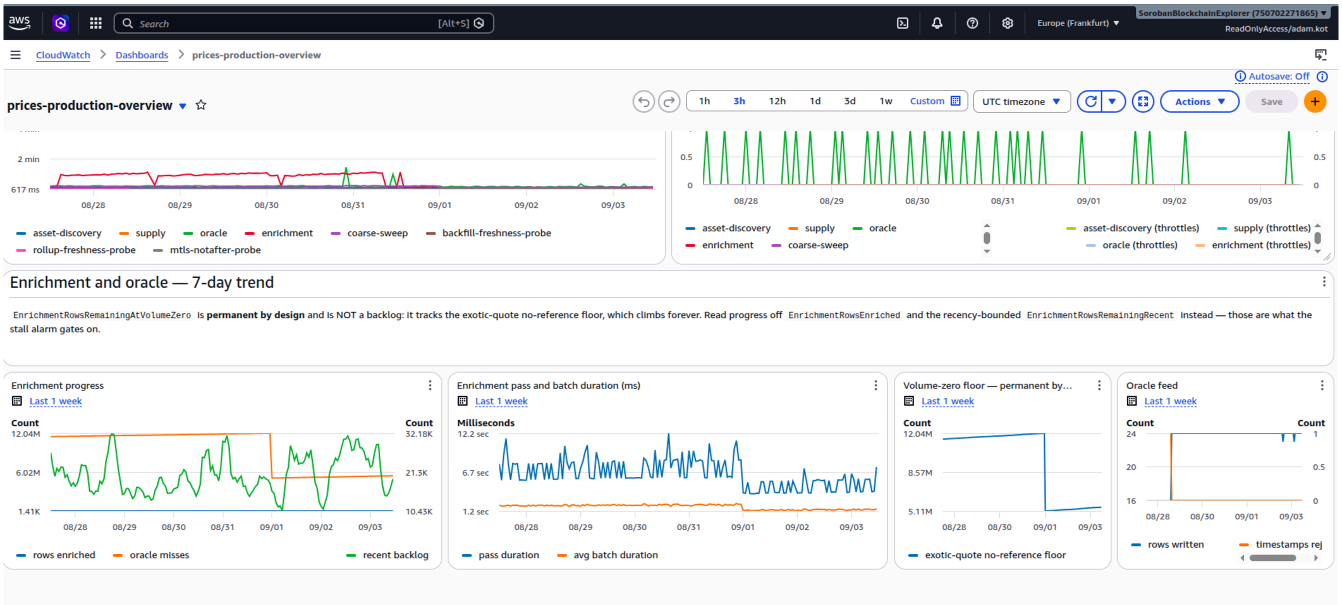
A synth-time assertion runs in CI so the dashboard cannot silently lose a widget.






Search
[Alt+S]
Europe (Frankfurt)
SorobanBlockchainExplorer (750702271865)
ReadOnlyAccess/adam.kot
CloudWatch > Dashboards > prices-production-overview
Autosave: Off
prices-production-overview
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Actions
Save
Ingestion
ClickHouseWriteLatencyMs is published by the ledger-processor itself (task 0125, Decision B2) and is empty until ComputeStack is deployed. It is timed at the candle-INSERT call site, so a write that retry_with_backoff retried folds its backoff sleeps in and reads as one long write.
Ingest queue — oldest message age (s)
Seconds
1 sec
500 ms
0 sec
10:00
11:00
12:00
oldest doorbell age
Ingest DLQ — messages visible
Count
1
0.5
0
10:00
10:30
11:00
11:30
12:00
12:30
DLQ depth
Ledger-processor — duration, errors, concurrency
Milliseconds
Count
397 ms
348.4 ms
299.8 ms
1
0.5
0
10:00
11:00
12:00
duration p95 (ms)
errors
concurrent executions
ClickHouse candle-write latency (ms)
Milliseconds
117.8 ms
91 ms
64.2 ms
10:00
11:00
12:00
p50
p95
Maximum
ClickHouse and backfill — 14-day trend
Named substitution (Decision A). The frozen SCF submission asks for "DB CPU". ADR 0007 replaced RDS with the shared Hetzner ClickHouse cluster, so that item is served here by the ClickHouse host and write-path metrics — free disk percentage plus ClickHouseWriteLatencyMs . Literal DB CPU is not readable: the runtime identities hold no grant on the ClickHouse system tables, and host CPU of a volume shared at a few percent says nothing about our write path.
ClickHouse host — free disk (%)
Last 2 weeks
Rollup lag per table (s)
Last 2 weeks
Backfill push age (s) — soroban_amm
Last 2 weeks
mTLS client cert — days to NotAfter
Last 2 weeks


Search
[Alt+S]
Europe (Frankfurt)
SorobanBlockchainExplorer (750702271865)
ReadOnlyAccess/adam.kot
CloudWatch > Dashboards > prices-production-overview
Autosave: Off
prices-production-overview
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Actions
Save
ClickHouse host — free disk (%)
Last 2 weeks
Percent
27.81%
25.53%
23.26%
08/22 08/24 08/26 08/28 08/30 09/01 09/03
free disk (%)
Rollup lag per table (s)
Last 2 weeks
Seconds
803.8 hr
401.9 hr
18 sec
08/22 08/24 08/26 08/28 08/30 09/01 09/03
price_ohlcv_1m
price_ohlcv_15m
price_ohlcv_1h
price_ohlcv_4h
price_ohlcv_1d
Backfill push age (s) — soroban_amm
Last 2 weeks
Seconds
1,217.9 hr
1,050.4 hr
882.9 hr
08/22 08/24 08/26 08/28 08/30 09/01 09/03
push age
mTLS client cert — days to NotAfter
Last 2 weeks
293
days to expiry
Scheduled workers — 7-day trend
The cleanup worker is deliberately absent: its EventBridge rule is disabled, so it publishes no Lambda metrics at all. It is still covered by the alarm strip.
Worker duration p95 (ms)
Last 1 week
Milliseconds
4 min
2 min
617 ms
08/28 08/29 08/30 08/31 09/01 09/02 09/03
asset-discovery
supply
oracle
enrichment
coarse-sweep
backfill-freshness-probe
Worker errors and throttles
Last 1 week
Count
Count
1
0.5
0
08/28 08/29 08/30 08/31 09/01 09/02 09/03
asset-discovery
supply
oracle
asset-discovery (throttles)
supply (throttles)



The alarm strip covers 49 alarms, derived from the construct tree rather than hard-coded, so the count maintains itself as alarms are added.

Cost, which the task was asked to record: roughly 450,000 metric publications a month at 0.01*perthousand*, *about***4.50, plus about $0.30** for the custom metric.

🔑 **One named substitution, stated in full.** The frozen Milestone 1 submission mentions "DB CPU". That is served here by ClickHouse host and write-path metrics, free disk percentage and write latency, because ADR 0007 replaced RDS with the shared Hetzner cluster. Literal database CPU is not readable anyway: our runtime identities hold no grant on the ClickHouse system tables, and host CPU of a volume we occupy at a few percent would say nothing about our write path.

A second substitution: the criterion says read-only IAM *role*; what is delivered is an IAM *user* with a narrow inline read policy, because there is no external principal to trust — the reviewing identity has not been named. A first draft granted the managed CloudWatch read-only policy, which would also have exposed **every log group and trace in an account shared with another project**. That was caught in review and replaced before merge.

⚠️ **Two open items found while building it, neither yet filed as a task.** The SDEX push-freshness alarm watches a metric series that has never existed, because the dimension it filters on has only ever been published by the other stream, so **a stalled SDEX backfill would not fire it** and it reports healthy because missing data does not breach. And the cleanup worker is deployed but deliberately dark, its schedule disabled after it destroyed the output of a repair campaign.

7.3 The API edge: CORS, custom domain, and the WAF decision

All three delivered.

The custom domain is live at `https://prices-api.sorobanscan.rumblefish.dev`, a regional custom domain mapped at the root, so no URL in this package carries a stage path.

CORS preflight is deployed on all seven `/v1` routes, permitting any origin without credentials. Verified on production and, separately, **in a real browser**: all seven routes returned 200 from cross-origin `fetch` with no CORS failure.

```
curl -X OPTIONS -i https://prices-api.sorobanscan.rumblefish.dev/v1/assets/native/price \
  -H 'Origin: https://example.com' -H 'Access-Control-Request-Method: GET'
# 204, access-control-allow-origin: *
```

A WAF was decided against, with the reasoning and the reversal triggers recorded. This is a public, read-only, key-gated API over public blockchain data with no personal data, no free-form input reaching a query, and rate abuse already bounded at two levels. A web ACL would cost $5 to $6 a month against a total infrastructure budget of about $108. The gap the throttles leave is per-caller *volume* rather than rate, and that is stated as the honest counter-argument. Four named triggers would reverse the decision, including any route that accepts free-form input or writes.

🔴 **Two failures during this work are disclosed, because both are more instructive than the result.**

The deploy took production down for about nine minutes. The compute deploy does not build the Rust binary, so it shipped a stale artifact whose handler panicked at cold start and every Lambda-backed route returned 502. **Every check in the runbook was structurally blind to it**, because the health probe and all seven preflights are gateway mocks that never reach the handler. The outage looked like a clean deploy.

The first cut of the CORS fix shipped half the mechanism. Adding preflight emits only the `OPTIONS` mock; on a proxy route the real response's allow-origin can only come from the handler, and the handler's CORS layer was scoped to a different route group. Preflight would have returned 204 and the browser would still have blocked the actual request, **with `curl` passing throughout** — which is precisely the defect the task existed to close, nearly re-shipped inside its own fix.

🔑 **A method note worth keeping.** The browser proof is that the `fetch` *resolved*, not that a header was read back. `Access-Control-Allow-Origin` is not a safelisted response header, so script cannot read it cross-origin at all. The first check script printed a null header on all seven passing routes, and that column was unreadable by construction. A verification script that reads a header the browser forbids will report a false negative on a working system.

8. What is deliberately not claimed

Milestone 2 is "Public API". The following are either Tranche 3 scope or known open work, and this submission does not claim them. They are listed so a reviewer can calibrate what "complete" means here — the same discipline Milestone 1 was accepted on.

Item	Status	Where it lands
Swagger **UI**	The OpenAPI **document** is served and rendered as an API reference (§7.1). An interactive Swagger UI is not deployed.	Tranche 3
Self-service onboarding portal	Key issuance is operator-run today, per a documented runbook.	Tranche 3
Integration suite running **in CI**	The conformance suite is operator-run against production and is deliberately not a CI job; a database service container in CI was decided against.	Tranche 3
Security review	Not performed.	Tranche 3
Public repository	The repository is private.	Tranche 3
7-day post-launch report	Not applicable until launch.	Tranche 3
Cache hit and miss percentiles reported separately	Hits measured. Misses are **not obtainable by the load-test method** — the system stops serving before a miss percentile can be sampled at 100 req/s (§5, AC 2).	Task 0260
Cold-start incidence and ClickHouse query time	Blocked on production CloudWatch access, not on the run.	Task 0260

Root cause of the read-path collapse under a zero-hit-rate load	Not established. The outage is reported in full in §5, AC 2.	Task 0260
USDC priced by measurement rather than assertion	Its entire series is peg-derived and returns `1` through a real depeg (§5, AC 6).	Task 0265, M3
The 2023-03 dislocation on two unrelated assets at an identical ratio	Real, reproducible, mechanism **not established**. The peg explanation is refuted.	Task 0266
`soroban_amm.earliest_data_available` overstates by 17 days	In the same reviewer-facing payload as AC 5.	Tasks 0263, 0264
Aquarius **concentrated** pool decoding	20 mainnet pools held back from the registry seed; their volume is absent from the source figure (§6.3).	Task 0080
Outlier filtering of the **headline price**	Deliberate gap: §7 scopes outlier detection to the VWAP. The asymmetry is documented (§6.2).	Task 0217
Dust-trade candles producing absurd closes	The volume threshold filters a **source**, not a **candle**.	Task 0116
Pagination cursor bound to its sort order	Switching sort mid-walk yields a wrong page (§6.4).	Task 0206
SDEX push-freshness alarm watches a series that has never existed	A stalled SDEX backfill would not fire it (§7.2).	Unfiled; see 0125
`info.license` emitted empty in the OpenAPI document	Licensing decision open.	Task 0155

Table — out-of-scope and known-open items, stated explicitly.

9. Live endpoints and access

Resource	URL / address	Access
Production API base	`https://prices-api.sorobanscan.rumblefish.dev`	`x-api-key`, key on request
OpenAPI document	`…/api-docs-json`	**Anonymous**
Health probe	`…/health`	Anonymous
Asset list	`…/v1/assets`	`x-api-key`
Asset detail	`…/v1/assets/{asset_identifier}`	`x-api-key`
Current price with `sources`	`…/v1/assets/{asset_identifier}/price`	`x-api-key`
OHLCV candles	`…/v1/assets/{asset_identifier}/ohlcv`	`x-api-key`
Batch prices	`POST …/v1/prices/batch`	`x-api-key`
Oracle cross-reference	`…/v1/oracles/{asset_identifier}`	`x-api-key`
Backfill status	`…/v1/backfill/status`	`x-api-key`
API reference (rendered)	`https://sorobanscan.rumblefish.dev/api/docs`	Anonymous
Production ClickHouse	`ch.sorobanscan.rumblefish.dev`, database `prices`	mTLS, client certificate on request
CloudWatch dashboard	`prices-production-overview`, `eu-central-1`	IAM, read-only viewer on request
Production alarms	`prices-production-*`, `eu-central-1`	IAM, read-only

GitHub repository	`https://github.com/rumblefishdev/stellar-prices-api`	Private during Tranche 2

Table — live verification endpoints and the access model for reviewers.

⚠️ **The `execute-api` URLs cited in the Milestone 1 package are retired.** That endpoint was switched off on 2026-09-02 and now answers `403`. The Milestone 1 documents were amended in place with dated notes rather than left pointing at a dead host, so every command in both packages remains runnable as written.

Reviewers wanting hands-on access to the key-gated or private resources — an API key, a short-lived mTLS client certificate, or the read-only dashboard viewer — can request them via the address on the submission form.

10. Repository navigation

Topic	Path
Technical design, including §9 criteria and revision history	`docs/prices-api-general-overview.md`
Deviations from the RFP and criteria wording	`docs/scf/milestone-2-rfp-deviations.md`
Cache verification, the full AC 3 measurement set	`docs/prices-api-cache-verification.md`
Backfill depth verification, AC 5 and AC 6	`docs/prices-api-backfill-depth-verification.md`
Load-test report, AC 2	`docs/prices-api-load-test-100rps.md`
Route, auth and TTL table of record	`docs/scf/api-endpoints.md`
Conformance suite, AC 1	`tools/scripts/conformance-0120.mjs`
Load-test script	`packages/prices-api/loadtest/price_load.js`
REST API (axum)	`packages/prices-api/`
Current-price materialised view, VWAP and outlier filter	`packages/prices-clickhouse/schema/current.sql`
ClickHouse schema, source of truth	`packages/prices-clickhouse/schema/init.sql`
AWS CDK app	`infra/`
Operator runbooks	`docs/runbooks/`
ADRs	`lore/2-adrs/`
Milestone 2 task ledger	`lore/1-tasks/archive/`, tag `milestone-M2`

Table — repository paths for implementation, infrastructure and decision review.

Key ADRs for Milestone 2 context:

- **ADR 0008** — single axum Lambda for the API
- **ADR 0011** — `base_currency` denominates rather than filters, and the unpriced bucket contract that AC 1's suite asserts against
- **ADR 0012** — the API Gateway stage cache stays; no CloudFront and no `X-Cache` header, with three alternatives closed and a when-to-revisit list